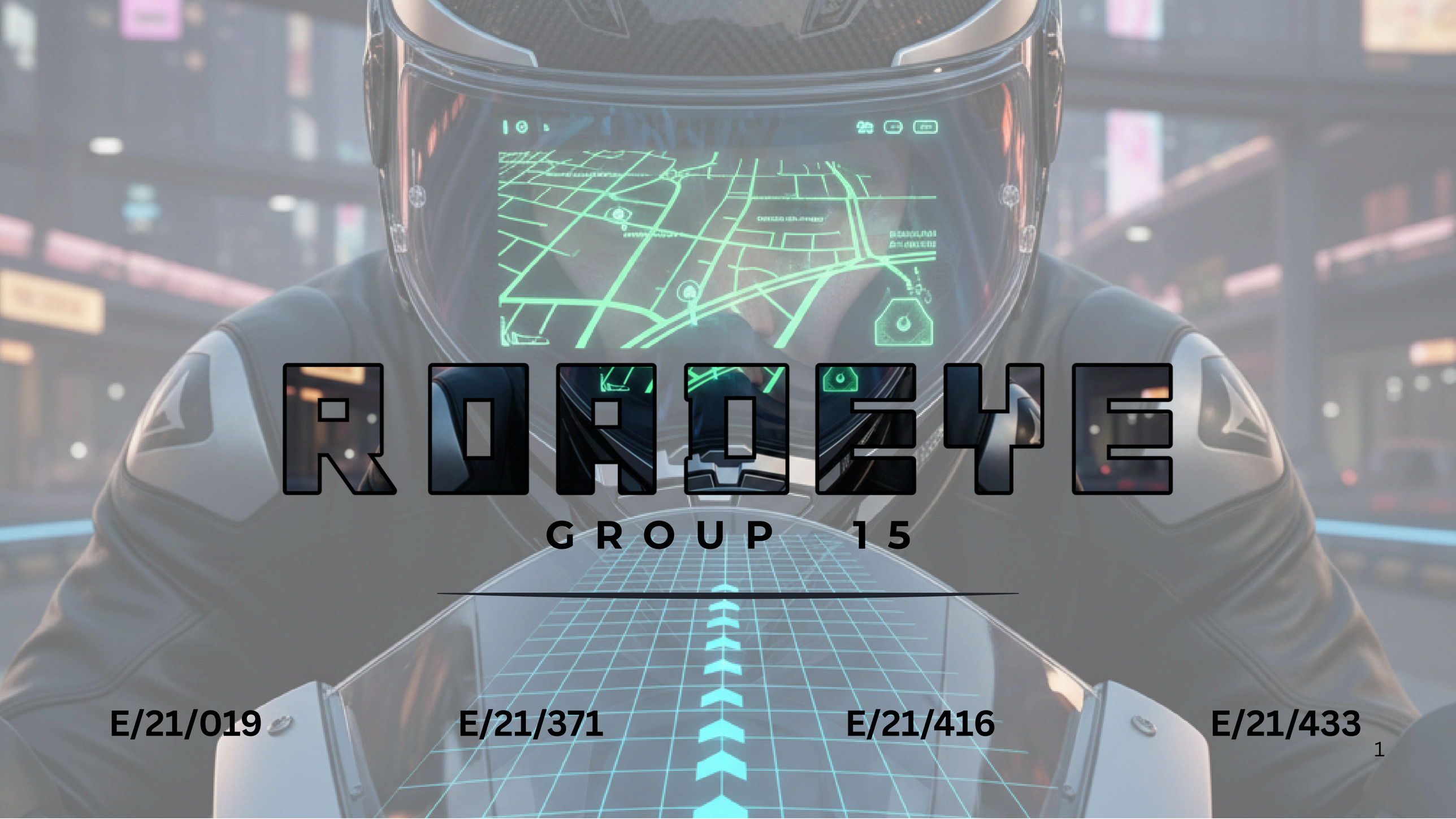


ROADEYE

GROUP 15

E/21/019 E/21/371 E/21/416 E/21/433

MOTIVATION

Motivation

- Motorcycle riders are vulnerable on the road,

  Why?

  - Limited situational awareness
  - Poor visibility in weather conditions
  - Rider distraction

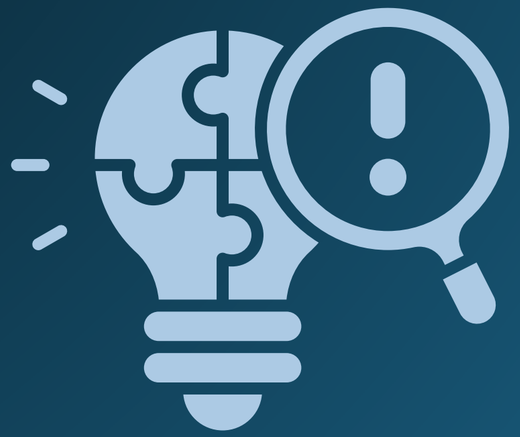
- Modern vehicles collect lot’s of data via sensors to assist the driver, motorcyclists only rely on vision

BACKGROUND

Why existing solutions are not enough

- Mirrors → require head-down attention
- Mobile phone navigation → causes visual and mental distraction
- Most helmets → lack active safety features

PROBLEM DEFINITION



Identified Problems

- Motorcycles helmets reduce field of view and introduce blind spots on the edges of vision
- riders lack options for communication and info
- Emergency response after crashes is often delayed

Core Challenge

How can we deliver critical safety information to a rider without increasing distraction or cognitive load?

PROJECT OBJECTIVES

Design Principles

- minimally distracting design
- Context-aware alerts
- Modular and scalable architecture
- Low power consumption

Main Objectives

- Enhance rider safety + awareness
- Hands-free interaction
- Minimize visual and mental distractions
- Detect situations before they become dangerous
- Emergency response and crash alerts

TARGET USERS & IMPACT

Target Users

- Daily motorcycle commuters
- Long-distance riders
- Delivery drivers

Expected Impact

- Reduced accident risk
- improved situational awareness
- Better riding habits
- faster emergency response

HIGH-LEVEL SYSTEM ARCHITECTURE

System Consists of 3 Components

- Helmet Unit - communicate with the driver
- Bike Module - gather info on the bike and the environment
- Mobile App - navigation, auxiliary sensors, store data

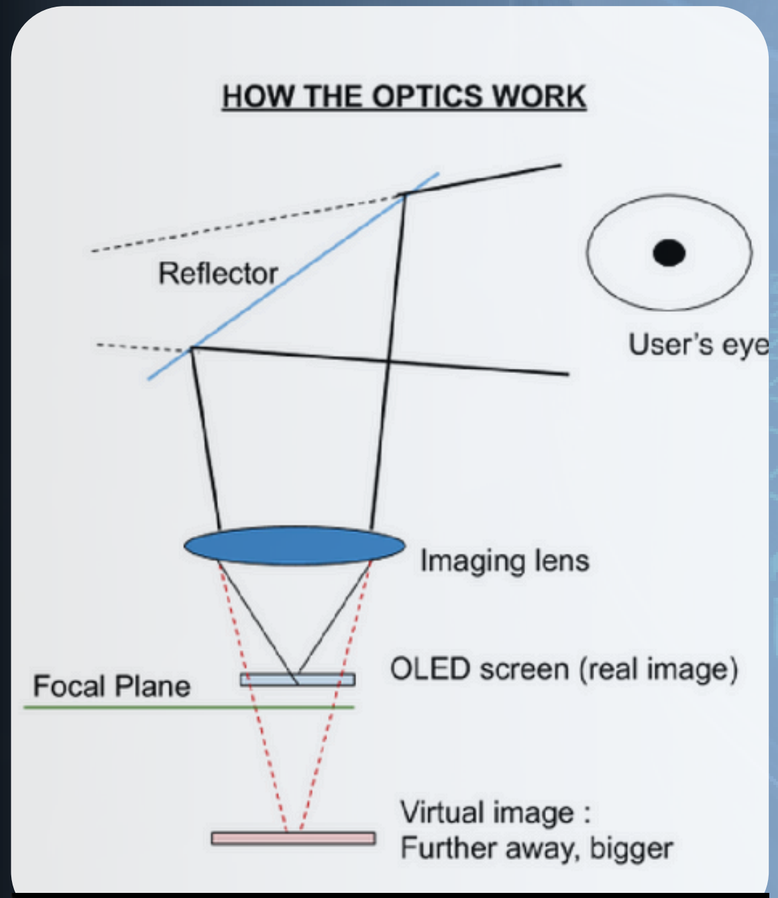
HELMET MODULE

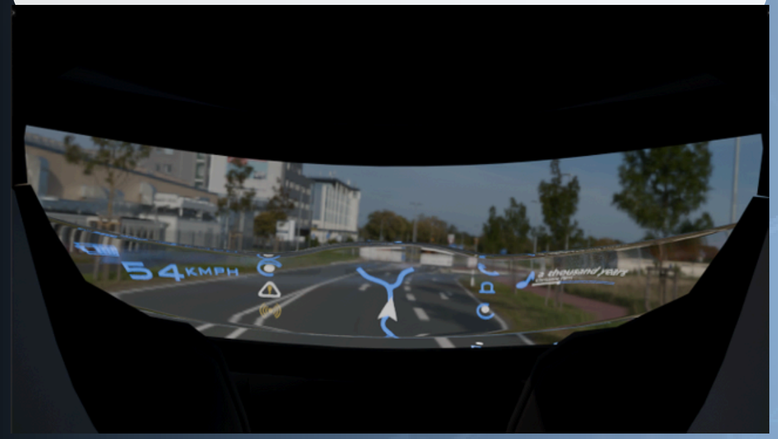
Communicating with user

- **Heads Up Display**
- **Stereo speakers**
  two speakers embedded in the cushion
- **Haptic motors**
  at the back of the riders ear

LED strip

- warn others on the road
- improve visibility of the rider during night time





HUD WORKING PRINCIPLE

Key Challenge

- Human eyes cannot safely focus on near displays while riding

Solution

- Fresnel lens + reflective combiner
- Creates a virtual image at a far distance
- Minimal eye refocusing
- Reduced eye strain & distraction

HELMET MODULE

- **Data collection**

- 9 axis IMU

- head motion, guestures, crash detection

- Hall effect sensor

- helmet buckle state detection

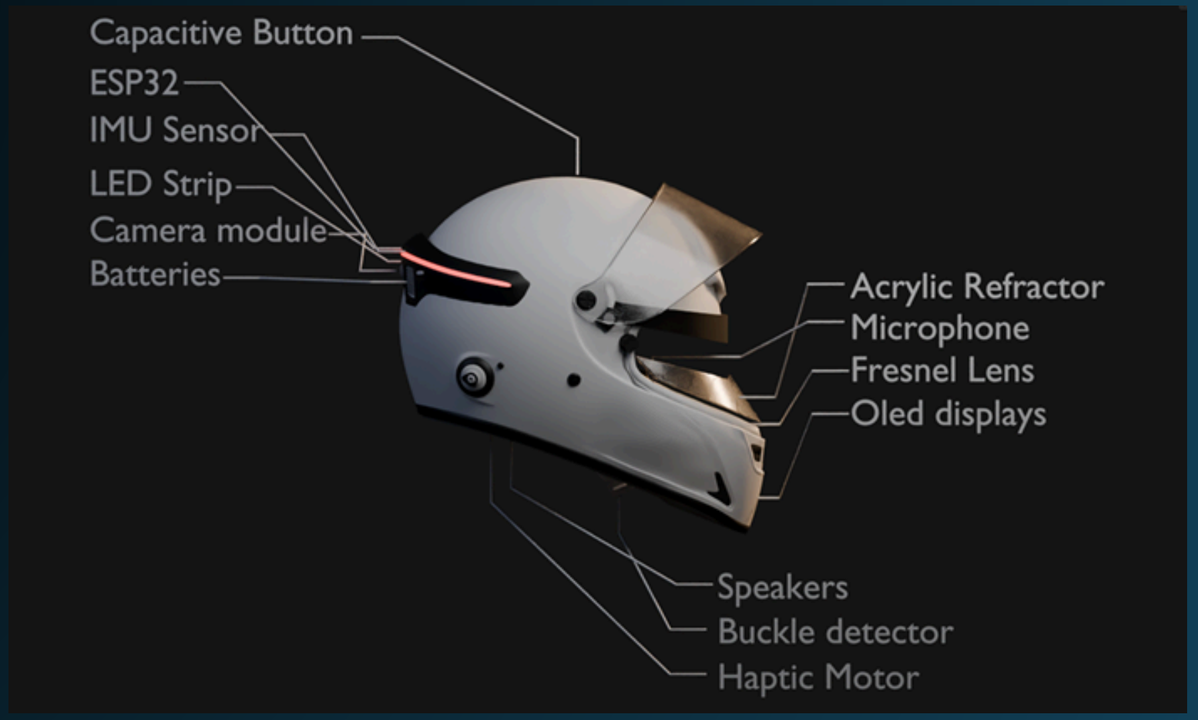
- Capacitive button

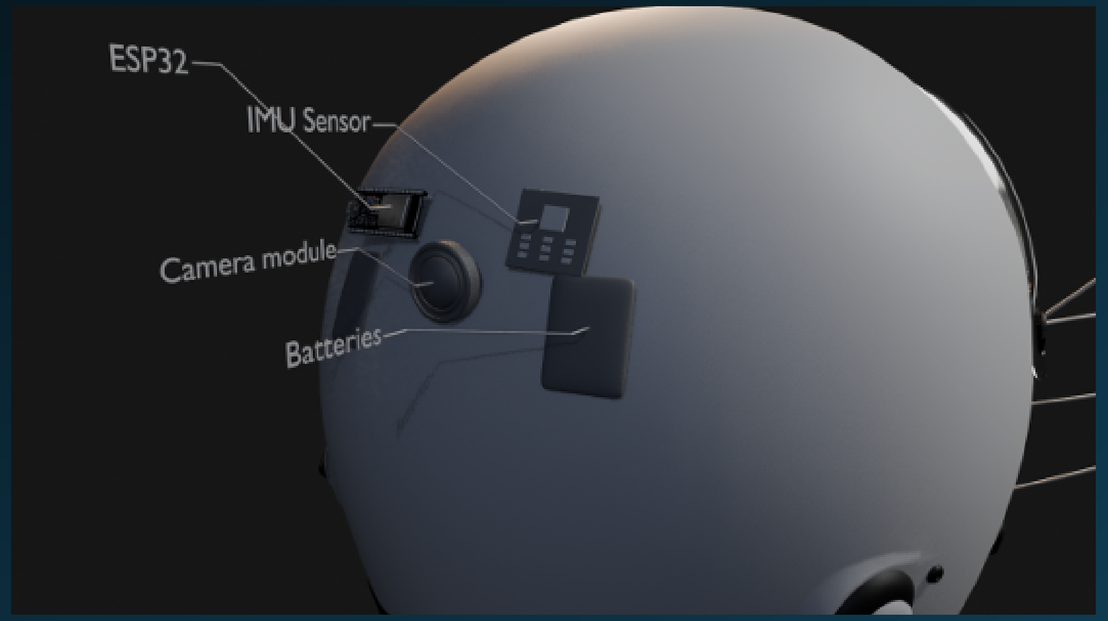
- wear detection

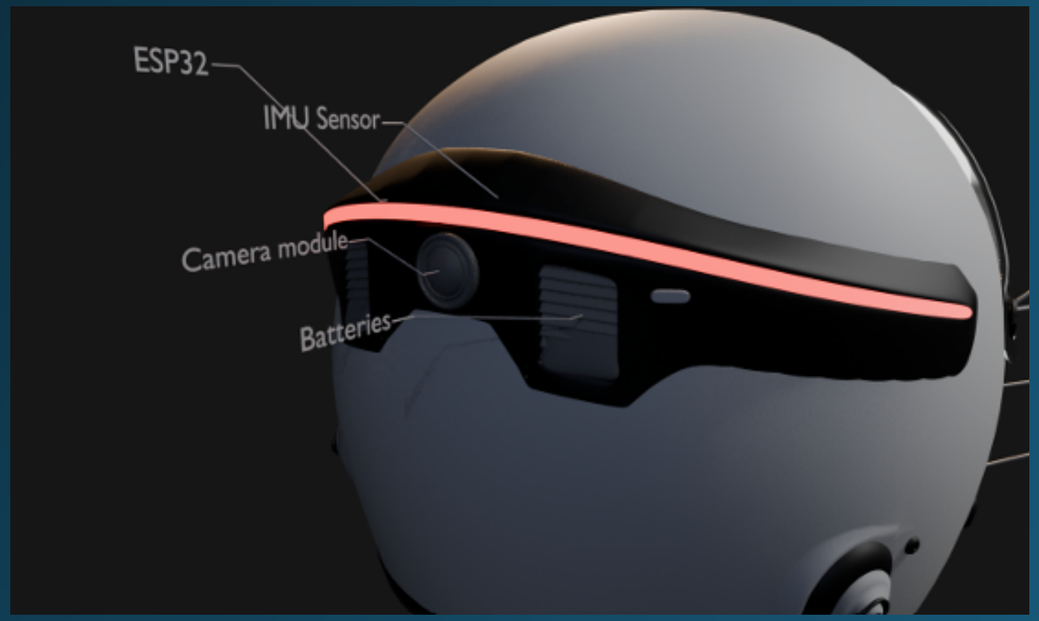
- Speaker

- rider voice input

- **Liop and Charging Circuit**


Capacitive Button
ESP32
IMU Sensor
LED Strip
Camera module
Batteries
Acrylic Refractor
Microphone
Fresnel Lens
Oled displays
Speakers
Buckle detector
Haptic Motor


ESP32
IMU Sensor
Camera module
Batteries


ESP32
IMU Sensor
Camera module
Batteries

BIKE MODULE

- **Majority of data is collected here**
- **Mounted at the back of the bike**

- 3 distance sensors
  - proximity warnings

- 9 axis IMU
  - acceleration, breaking, tilt and crash detection

- Humidity, Altitude, pressure, Temperature, ambient light sensors
  - Fusion with weather data, environmental conditions

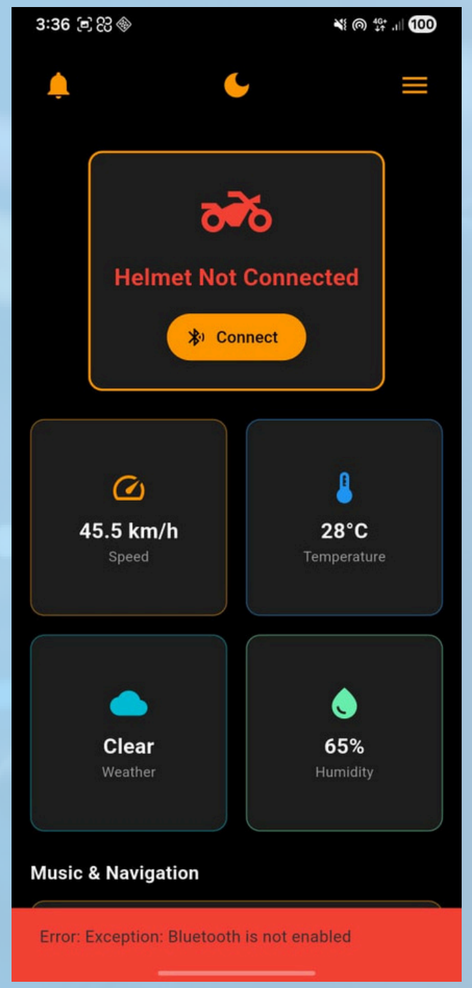
- Vibration sensor
  - road conditions

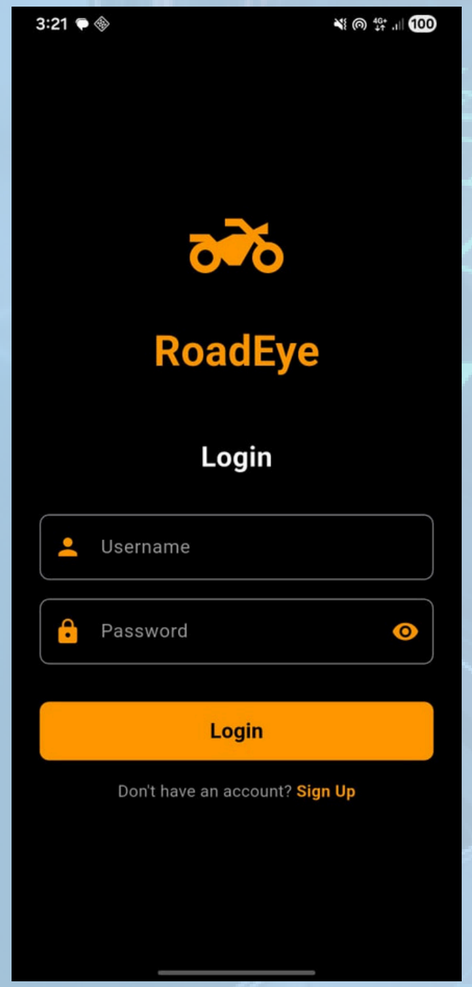
- **Liop and Charging Circuit**

MOBILE APPLICATION

- Navigation data (simplified view)
- Traffic and weather alerts
- User selected notifications
- forward phone calls and music playback
- store data collected from sensors
  - analyze data, creates a summary to be viewed later
- emergency contacts
- update and configure other two modules

MOBILE APPLICATION





App Functions

- Navigation & traffic
- Weather alerts
- Smart notifications
- Ride analytics & history
- Emergency contact alerts
- Firmware updates (OTA)

Role

- High-level intelligence
- User customization
- Data visualization

DATA GATHERING AND PRESENTING

- **Navigation** - GPS through mobile app
  - shows a simplified map
- **Upcoming weather** - what to expect further ahead
- **Speed estimate** - Fusion of GPS and IMU readings

DATA GATHERING AND PRESENTING

- **Rear and side collision warnings**
- **Rain, Fog, Wet road conditions**
  - Using Humidity + weather alerts
  - Dynamically reduce threshold for collision warnings
- **Night time riding**
  - Change HUD brightness
  - Dynamically reduce collision thresholds

DATA GATHERING AND PRESENTING

- **Riding habbits**
  - Aggressive acceleration, breaking and tilting

- **Road conditions**
  - Adjust riding habit thresholds accordingly

DATA GATHERING AND PRESENTING

Mobile connection

- **Answer phone calls, Play music**
  - gestures to interact (accept calls, play/pause music)
- **Helmet to helmet communication**
  - When there’s a passanger
- **Display notifications and messages**
  - forward notifications from selected apps to the helmet

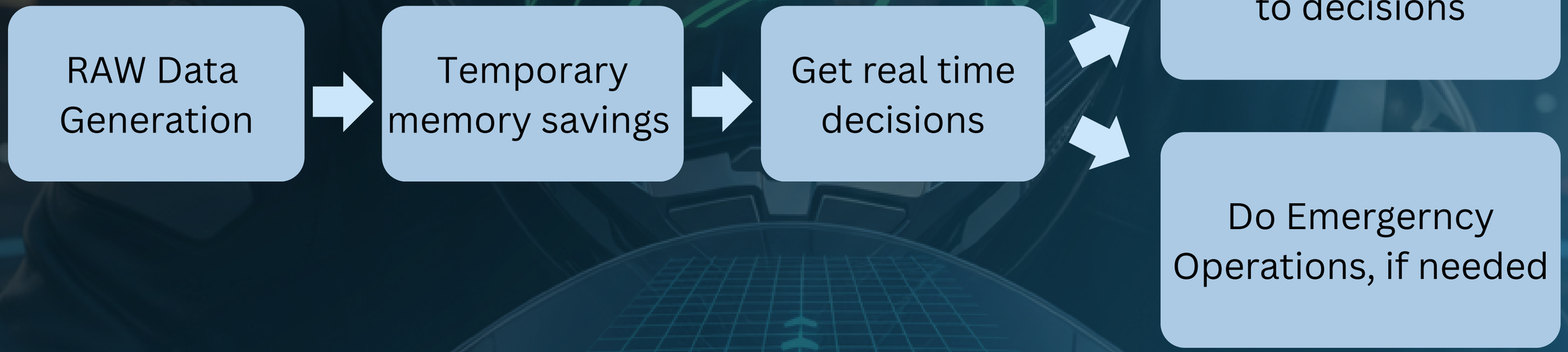
DATA GATHERING AND PRESENTING

- **Anti theft protection**
  - detect tampering while in parking mode
- **Helmet buckle reminder**
  - alert the rider if the buckle is not clipped in
- **Crash detection**
  - alert preselected contacts in case of emergancy

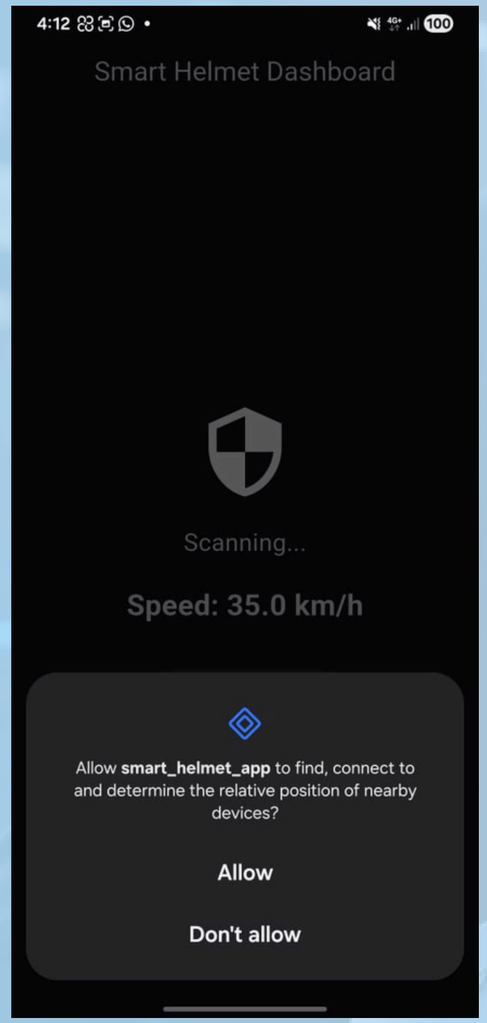
COMMUNICATION BETWEEN MODULES

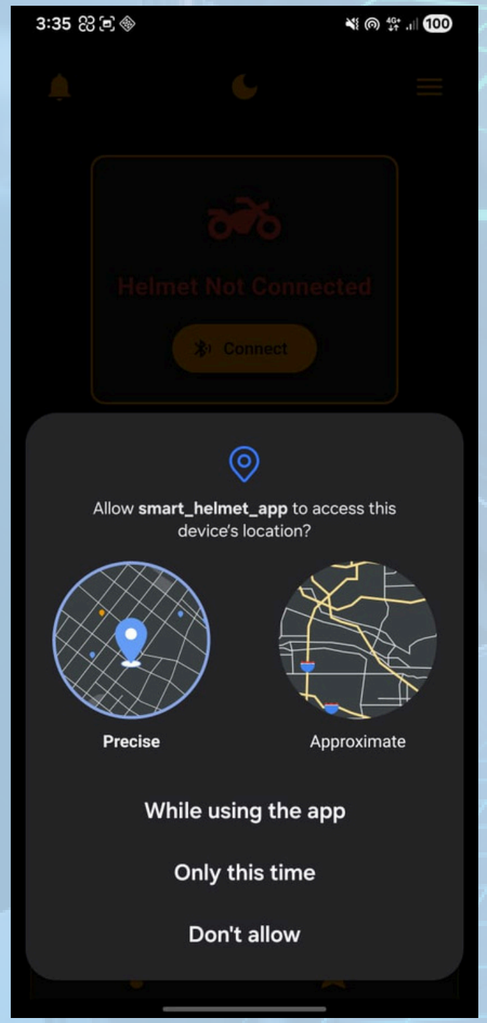
- Helmet and bike module connected via wifi
- Helmet and mobile app connected via bluetooth
- Potential for two helmets to connect over wifi for helmet to helmet communication

FLOW ARCHITECTURE



SECURITY & PRIVACY CONSIDERATIONS





Security

- Bluetooth pairing authentication
- Secure firmware updates (OTA)
- Device-to-device encrypted communication

Privacy

- User controls what data is stored
- No continuous cloud tracking
- Emergency data shared only during crashes

LEGALITY AN RIDER SAFETY

- **Motor Traffic (Amendment) Act, No. 10 of 2019**

"152A. (1) No person shall use a telephone, mobile phone or any other **movable communication** equipment when driving a motor vehicle on a road."

- By law it is prohibited to operate any handheld device and give it attention while driving.
- Our design makes sure that answering phone calls does not require any hand operation and does not obstruct riders hearing like ear buds (speakers embedded into the cussion)

LEGALITY AN RIDER SAFETY

- **Motor Traffic Act**

  “152. (1) No person shall drive a motor vehicle on a road when he is in such a position that he **has not or cannot have full control of the vehicle** or a **full and unobstructed view** of the road and the traffic ahead of the motor vehicle.”

- refracter of our heads up display is made or clear acrylic so that it does not cause any obstruction to the riders view
- Our system does not require any hand input form the user so the rider always have full control of the vehicla

LEGALITY AN RIDER SAFETY

- **SLS 517:1994 standard mandates;**

A hard outer shell, A shock-absorbing liner, A chin strap. Buffer padding. A head band

- our system does not obstruct the above required construction of the helmet in any way
- we only introduce materials that are recognized by SLS 517 as safe (no glass or metal)

BUDGET ESTIMATE

Module	Price (Rs)
ESP32-C3-MINI-1 microcontroller	1200
Sensors	9800
LiPo and charging circuit , wires	4500
OLED displays	1180
Camera model, Microphone & Speakers	1800
Fresnel Lense	1500

Estimated Budget

LKR 20,000

ROADMAP DEVELOPMENT PHASES

Weeks 1–2
Research & final design

Weeks 3-5
Hardware procurement

Weeks 6-9
Helmet module development

Weeks 10-12
Bike module development

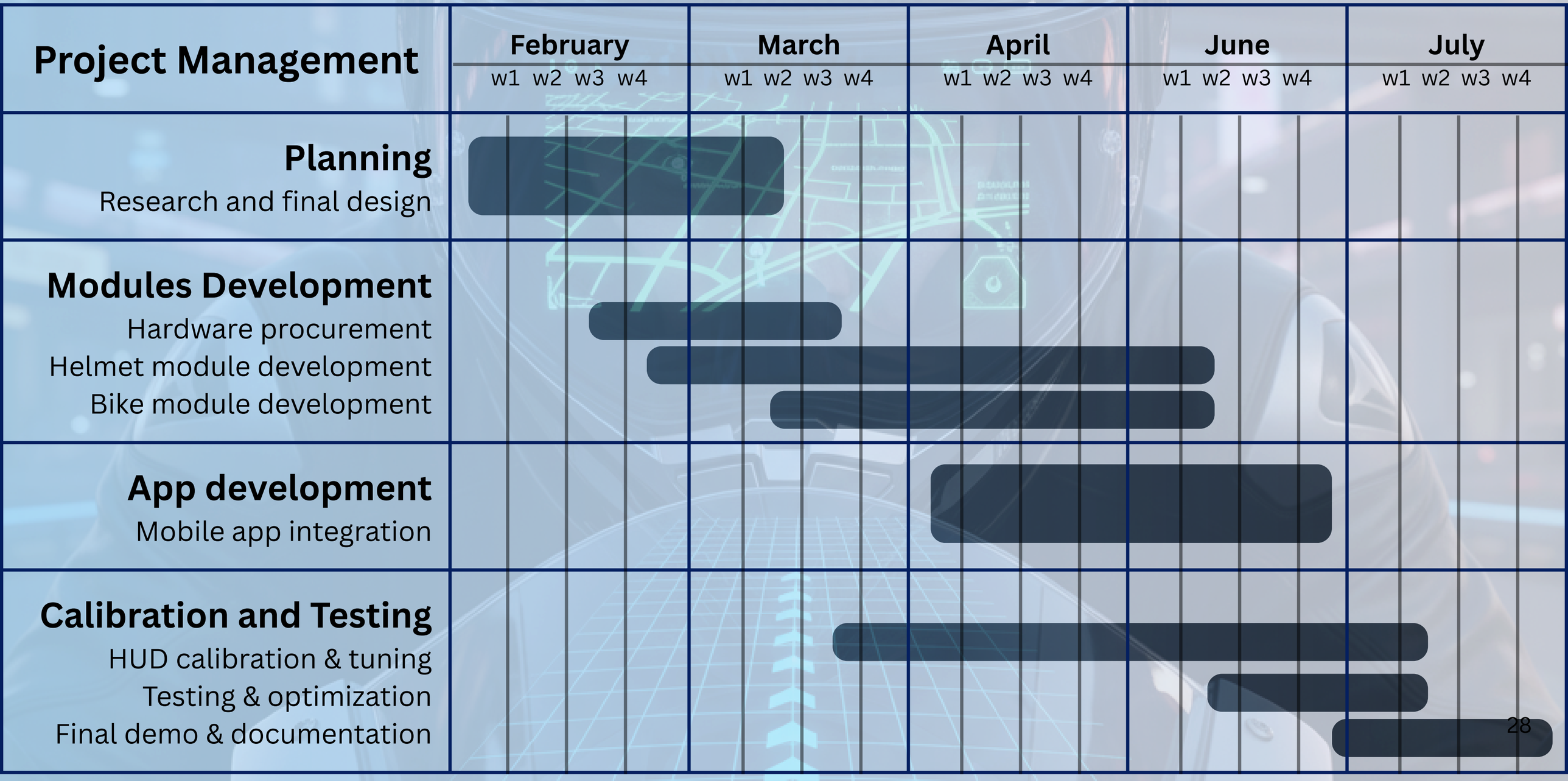
Weeks 13-17
Mobile app integration

Weeks 16–17
HUD calibration & tuning

Weeks 18–19
Testing & optimization

Week 20
Final demo & documentation

ROADMAP DEVELOPMENT PHASES



THANK YOU